Fraud Transaction Detection and Coordinated-Group Discovery

PaySim offline prototype | Analytics and Systems of Big Data | October 2026

Executive summary

This project evaluates two distinct tasks: transaction-level fraud classification and label-free discovery of candidate coordinated groups. XGBoost reached **0.971 test average precision** with pre-transaction balances, but the balance-free ablation fell to **0.326**. The high result reflects a strong PaySim-specific signature. The full graph scan returned **zero groups** meeting the declared coordination criteria. PaySim has transaction labels, not verified ring labels, so no PaySim ring-level performance is claimed.

Dataset and split

Transactions	6,362,620	**Fraud labels**	8,213 (0.1291%)
Steps	1-743	**Nulls / duplicates**	0 / 0*
Train	step <= 520	**Validation**	521-631
Test	step >= 632	**Test fraud prevalence**	1.399%

No missing fields were found; exact duplicates were checked by 64-bit row hash. Fraud labels occur only on CASH_OUT (4,116) and TRANSFER (4,097). Test prevalence is 14.7x training prevalence (0.095%), indicating substantial temporal shift. *A theoretical hash collision is possible.

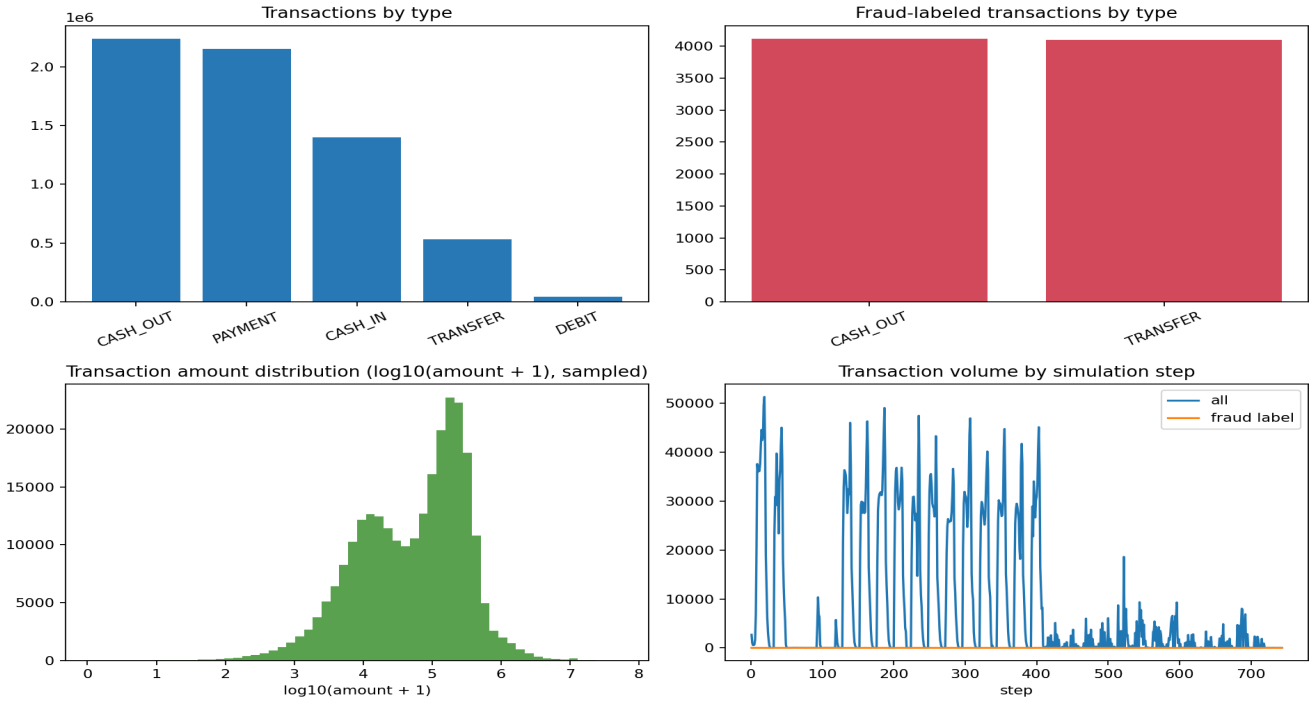


Figure 1. Transaction mix, fraud counts, log-scale amount sample, and activity over simulation steps. The account activity panel is based on a deterministic sample.

1. Feature construction and evaluation protocol

The transaction task predicts the PaySim label for each event. Account histories use only transactions from strictly earlier simulation steps. Records at the same step do not update one another because PaySim provides no within-step event order. Account IDs are used to construct histories but excluded from classifier inputs; fraud labels are never aggregated into account features.

Set	Predictors	Purpose
A	Type, step, amount, pre-event source/destination balances	Transaction-only reference
A without balances	Type, step, amount	Tests balance-signature dependence
B	A + prior endpoint counts, amount summaries, last-activity gap	Adds account behavior
C	B + prior distinct counterparties and repeat intensity	Adds simple graph neighborhoods

For a transaction at step s, account snapshots end at step s-1. The current event and all other step-s events are unavailable to its history features. This gives a prospective offline feature definition without inventing within-step order.

Chronological evaluation

Split	Rows	Fraud	Rate	Purpose
Train through 520	6,082,007	5,781	0.095%	Fit models and class weights
Validation 521-631	191,147	1,180	0.617%	Early stopping / selection
Test from 632	89,466	1,252	1.399%	Final evaluation

Test fraud prevalence is 14.7 times the training prevalence. The chronological split preserves detection direction, but the shift means test metrics describe this final PaySim period rather than a stationary deployment population.

Model fitting and metrics

Logistic Regression uses class weighting. XGBoost uses histogram trees, learning rate 0.08, depth 6, minimum child weight 10, row and column subsampling 0.8, and L2 regularization 5. The training-only positive-class weight is 1,051.07. Validation average precision controls early stopping (30 rounds; 600-tree maximum); test metrics are computed after model selection. The transparent rule baseline ranks transfer and cash-out events by log amount.

Average precision summarizes precision-recall ranking and is more useful than accuracy under severe imbalance; an all-negative classifier would have high accuracy. Operational results use exact top-k budgets of 90, 448, and 895 test alerts. Stable row order resolves score ties, and inclusive-cutoff overflow is recorded as a diagnostic.

The transaction-only model A reached the 600-tree ceiling rather than early-stopping; treat its score as the tested-cap result, not a confirmed optimum. The no-balance ablation stopped at two rounds; B and C stopped at 126 and 312.

2. Transaction fraud classification

Model A uses transaction type, amount, step, and pre-transaction origin/destination balances. The no-balance ablation removes both balance fields. Model B adds strictly prior-step endpoint behavior; Model C adds prior graph degree, distinct-counterparty counts, and repeat intensity. Post-transaction balances, `isFlaggedFraud`, raw account IDs, and target-derived features are excluded. Logistic Regression is the conventional baseline; XGBoost uses histogram trees, training-derived class weighting, validation AP for early stopping, and a 600-tree cap.

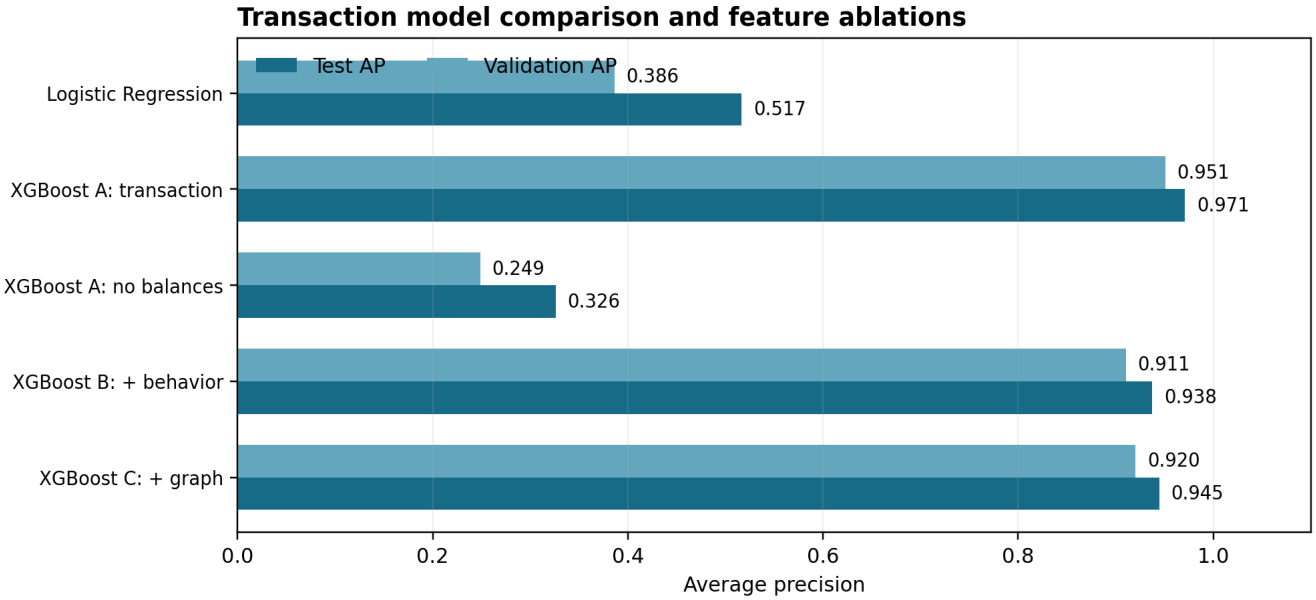


Figure 2. Validation and test average precision by model and feature set. The holdout is chronological; test ranking is evaluated once after validation-based model selection.

Model	Val AP	Test AP	P / R @ 90
Rule: transfer/cash-out amount	-	.218	.789 / .057
Logistic Regression A	.386	.517	1.000 / .072
XGBoost A	.951	.971	1.000 / .072
A, no balances	.249	.326	1.000 / .072
XGBoost B, behavior	.911	.938	1.000 / .072
XGBoost C, graph	.920	.945	1.000 / .072

Key observation: 8,034/8,213 fraud rows (97.82%) have amount equal to oldbalanceOrg, versus 0/6,354,407 non-fraud rows. The fields are available before the transaction, so this is not post-event leakage; it is a simulator-specific signal. AP falls sharply when balances are removed. Behavior reduces AP from A; graph features recover only about +0.007 AP over B. XGBoost A reached the 600-tree cap and is not confirmed converged.

At 90 alerts, perfect precision covers only 90 of 1,252 test frauds (7.19% recall). Fixed-budget metrics use stable top-k selection; a score threshold alone may exceed the budget when scores tie.

3. Operating points and interpretation

Average precision summarizes ranking across thresholds. The alert-budget view answers the operational question: how do precision and recall change when investigators can review only a fixed number of alerts? Labels are used here only for retrospective evaluation.

XGBoost variant	90 alerts P / R	448 alerts P / R	895 alerts P / R
A: transaction	1.000 / 0.072	1.000 / 0.358	0.990 / 0.708
A: no balances	1.000 / 0.072	0.616 / 0.220	0.434 / 0.310
B: behavior	1.000 / 0.072	1.000 / 0.358	0.984 / 0.704
C: graph	1.000 / 0.072	1.000 / 0.358	0.989 / 0.707

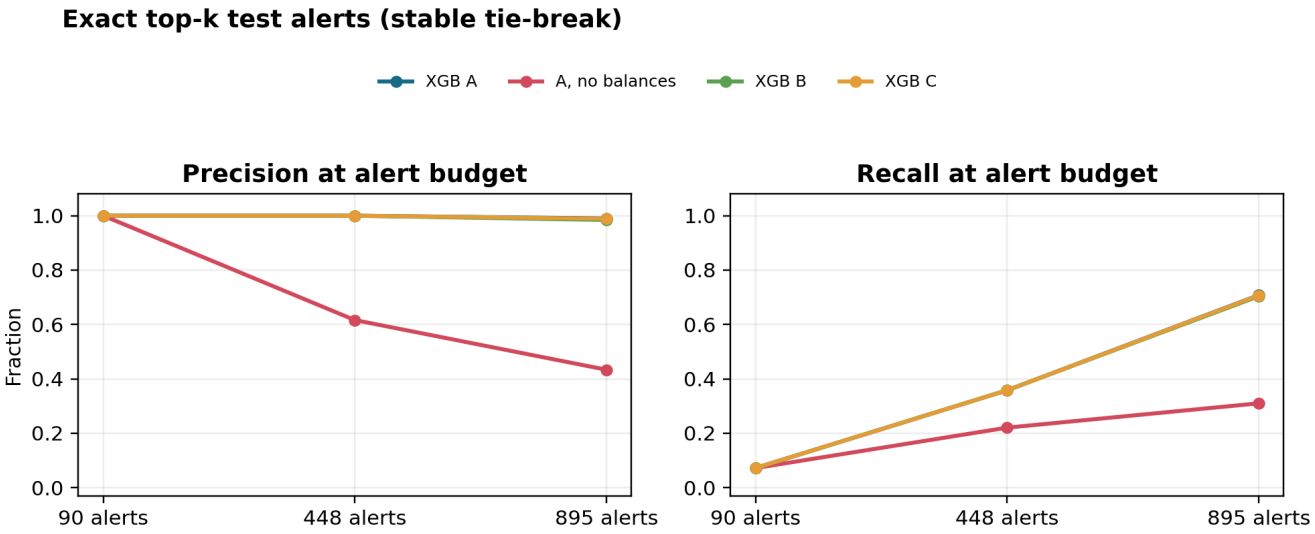


Figure 3. Precision and recall at exact 0.1%, 0.5%, and 1.0% test alert budgets.

Reading the operating points

At 90 alerts, all XGBoost variants retrieve the same 90 frauds and no false positives, giving 7.19% recall. At 448 alerts, A, B, and C each recover 448 frauds; the no-balance model recovers 276. At 895 alerts, A, B, and C remain close: precision ranges from 0.984 to 0.990 and recall from 0.704 to 0.708. The graph addition improves B by only 0.0074 test AP and does not surpass A.

Scores are tied near several cutoffs. At 90 alerts, an inclusive threshold would select 93 A rows, 214 no-balance rows, 199 B rows, and 165 C rows. The reported results select exactly k rows using a stable tie-break. Operational use would need a secondary ranking or queue policy for ties.

4. Coordinated-group discovery

The group miner is independent of transaction labels and model predictions. It scans non-overlapping 24-step windows and proposes weak components, strongly connected components, fan-in/fan-out motifs, and Charikar-style dense subsets (bounded to small components). Candidates require at least three accounts in a directed cycle or at least two accounts with temporally ordered relay activity across multiple steps. A fixed heuristic ranks structural density, cycles, temporal coordination, repeated pairs, flow continuity, and amount similarity; it is not a calibrated probability.

Full PaySim result: 31 windows scanned; 0 eligible groups. The earlier permissive screen returned chain-like components that failed the stricter coordination rule. This negative scan result does not show that PaySim contains no rings. The dataset has no independently verified ring IDs, so ring precision and recall on PaySim are unavailable.

Controlled ring-recovery check

Ranking	AP (mean +/- SD)	Precision@30	Member recovery@30	Ring recall@30
Evidence score	.241 +/- .040	.447 +/- .069	.357 +/- .055	.447 +/- .069
Transaction-count baseline	.201 +/- .004	0.000	0.000	0.000
Density baseline	.097 +/- .000	0.000	0.000	0.000

Benchmark: five seeds, each with 30 planted cyclic or relay groups and 90 decoy episodes. Matching is one-to-one at account-set IoU >= 0.5. Mean candidate coverage was 1.00, but top-30 recovery was modest. The synthetic background is easier than real transaction networks and does not validate PaySim ring accuracy.

How candidate groups are formed

Stage	Evidence / rule
Proposal	Weak components bound membership; SCCs, fan-in/fan-out, and dense peeling add subgraph proposals.
Coordination	Directed cycles, ordered inflow then outflow, temporal compactness, repeated pairs, amount similarity, and flow continuity.
Eligibility	At least 3 accounts in a directed cycle, or at least 2 temporally ordered relays over multiple active steps.
Ranking	Fixed weighted score for review priority; not a probability and not trained on PaySim fraud labels.

The strict screen favors coordination over mere connectivity, but can miss longer or cross-window rings, weakly connected groups, and partial activity. Windows do not overlap, and dense-subgraph peeling is limited to components of at most 20 accounts. These bounds control runtime but create false-negative risk.

Conclusions and next work

The strongest fraud-classification result is heavily tied to a PaySim balance signature; the balance-free result is substantially weaker. Behavioral and graph features do not improve AP over transaction-only XGBoost. No verified fraud ring was identified in PaySim. The controlled experiment supports continued development of interpretable group mining, but stronger noisy, overlapping, and cross-window injections plus sensitivity analysis are needed. Additional chronological cutoffs and independently labeled real data are needed for generalization claims. Streaming is deferred until offline ring recovery is validated.

Reproduction: commands are in README.md; metrics and ring outputs are in outputs/. Runtime / peak RSS: classifier 398 s / 7.3 GiB; group scan 276 s / 734 MiB.